# INTERIM FEDERAL SPECIFICATION

## BRACE, FLOOR SWEEPING BRUSH

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to preparing activity at the address shown above.

### 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of metal braces to be used with the type II floor sweeping brushes specified in H-B-651.

1.2 Classification.

1.2.1 Types. Floor sweeping brush braces shall be one of the following types, as specified (see 6.2).

Type I - Shield type.
Type II - Strut type.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-B-651 - Brush, Floor Sweeping.
PPP-B-636 - Box, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample brace covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Standard product. The braces delivered under this specification shall be the manufacturer's standard commercial product except for any design modification and additional requirements specified herein.

3.3 Materials. The materials used in the manufacture of the braces shall be of the same quality metal (see 3.3.1) found to be suitable for the purpose in commercial practice, unless otherwise specified.

3.3.1 Brace material. The brace shall be constructed from either commercial steel or brass in accordance with the contractor's standard commercial practice.

3.3.2 Protective plating. Braces manufactured from ferrous metal shall be plated with either cadium, chromium, zinc (galvanize), or similar plating and shall show no signs of corrosion when tested in accordance with 4.4.1.

3.4 Design. The braces shall be of a commercial design that has been use-tested and accepted by commercial users for strengthening and securely locking the handle into the brush block during use.

3.4.1 Type I brace. The type I brace shall be a reusable sheet-metal (see 3.3.1) type and shall be designed to be attached to the brush block and handle without the use of tools, drilling, or the securing of fastening devices into the brush block or handle (see figures 1A and 1B). The brace shall be secured by clamping, such as a "U" bolt and wing nuts, or by use of a spring like tension device. The design shall permit the user to easily interchange the handle between the brush block handle holes to equalize brush face wear. The brace, brush block, or handle shall show no signs of breaking, distortion, or loosening when tested as specified in 4.4.3.

3.4.2 Type II brace. The type II brace shall be a reusable strap-metal type having a loop or ring for the handle and two (2) supporting struts of equal length for attaching to the back of the brush block. The brace shall be designed so that it can be attached with hand tools that do not require the services of a skilled mechanic. The user shall be able to insert the handle through the loop or ring and thread it into the brush block handle hole after the brace is assembled to the block. Also, when the brace is reversed and reattached to the block using the same fastening holes, it shall be so positioned as to permit the handle to be threaded into the opposite brush block hole. The brace, brush block, or handle shall not show any signs of breaking, distortion, or loosening when tested as specified in 4.4.3.

3.5 Instructions. Complete instructions, fully describing the proper method of attaching the brace to the brush block and handle shall be furnished with each type I and II brace in accordance with standard commercial practice. The instructions for the type II brace shall include such information as: recommended drill sizes to be used, depth holes should be drilled, and distances holes should be located from a specified location. The braces shall be assembly tested to determine they fit properly without interference or misalignment when tested as specified in 4.4.2.

3.6 Workmanship. Workmanship shall be in accordance with the best commercial practice. Threads, when applicable, shall be of the National Form and shall be furnished with adequate means for preventing loss of tightness. Rivets, when applicable, shall be firmly and neatly set. All braces shall be free of sharp edges, burrs, slivers, rust, rough surfaces, areas with thin or missing plating, and any defects which may affect durability, serviceability or appearance.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for the defects in table I. The preproduction sample shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 End item inspection. The lot shall consist of all the same type braces offered for inspection at one time. The sample unit shall be one complete brace assembly.

4.3.1.1 Visual examination. Examination shall be made of the end item for the defects in table I. The inspection level shall be level II with AQL of 4.0 expressed in terms of defects per hundred units.

TABLE I. Visual defects.

| Examine | Defect |
|---|---|
| Finish | Not corrosion protective plated or made from corrosion resistant material. |
| Construction and Workmanship | Poor construction or workmanship affecting serviceability, durability, or appearance. Braces not free of sharp edges, burrs, or rough surfaces. Braces not clean, evidence of oil grease, or dirt. Plating thin or missing. Threads (if applicable) not cleanly cut, filled with plating or not National Form. Instructions missing or not legible. Rivets (if applicable) not neatly and firmly set. Components missing. |

4.3.1.2 Testing of the end item. The end item shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE II. Classification or preparation for delivery defects.

| Examine | Defect |
|---|---|
| Markings (exterior) | Omitted, incorrect, illegible, improper size, location sequence, or method of application. |
| Materials | Any component missing or damaged. |
| Workmanship | Inadequate application of components, such as incomplete closure of container flaps, loose strapping, or distortion of container. |
| Contents (interior) | Number per container is more or less than required. Net weight exceeds requirements. |

4.4 Test.

4.4.1 Corrosion test. Complete braces when fabricated of ferrous metal shall be placed in a boiling 10 percent (by weight) aqueous solution of sodium chloride for a period of not less than 15 minutes. The brace assembly upon being removed from the solution shall be immediately immersed for at least 1 hour in 10 percent (by weight) aqueous solution of sodium chloride at room temperature of 60 to 85 degrees F. Assembly shall then be removed from this solution and, without having the adhering liquid wiped off, allowed to dry for 24 hours at room temperature (68° to 85° F.) and then wiped free of any residue and examined for presence of corrosion. Evidence of corrosion shall be cause for rejection.



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4.4.2 Assembly test. The type I or II brace shall be assembled, according to the manufacturer's instructions furnished with each brace, using a finished brush block or brushhead (less bristles) and a handle meeting the requirements specified in H-B-651 to determine compliance with 3.5. Each brace passing the initial assembly test shall again be tested for complicance with 3.5 by being assembled in the opposite brush block handle hole. The failure of the brace to assemble in accordance with instructions shall be cause for rejection.

4.4.3 Weight test. A type I or a type II brace, as applicable, assembled and tested as specified in 4.4.2 shall be used for this test. The block shall be securely mounted in a vertical position in a suitable holding device with the handle in a horizontal position. Both the handle and brace shall be free from all interference and support during the test. Apply a steady force of 12 plus 1 pound by using a scale or weight in a downward direction for 30 plus 5 seconds, 48 plus or minus 1 inch from the block to determine compliance with either 3.4.1 or 3.4.2, as applicable. The assembly shall then be turned 180° and the test repeated.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A and B. Each type II brace, with hardware and instruction sheet shall be packaged in a box of the type normally used by industry. Type I braces requiring fastening hardware, such as U-bolt and wing nuts, shall be packaged in a envelope or box of the type normally used by industry.

5.1.2 Level C. Braces shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Braces in quantities as specified shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve braces shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. Braces shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the unit and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the unit and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The braces covered by this specification are intended to reinforce and securely hold the handle in the floor sweeping brushes covered in H-B-651.

6.1.1 Type I. For use with all size brush blocks and suitable for all general sweeping conditions requiring only moderate bracing in the brush block and handle area.

6.1.2 Type II. For use with 18 inch or longer brush blocks and intended for heavy duty sweeping conditions requiring more than moderate bracing in the brush block and handle area.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specify type required (see 1.2).
(c) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(d) When preproduction sample is required (see 3.1).



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6.3 Illustrations. The illustrations shown herein as figures 1A, 1B, and 2 are for the purpose of illustrating the general design and are not intended to preclude any braces which otherwise meet the requirements of this specification.

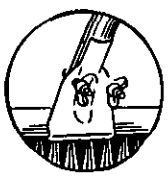


Figure 1A
"U" bolt type

Type I

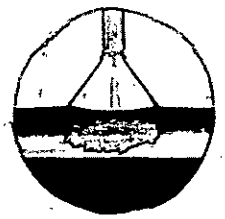
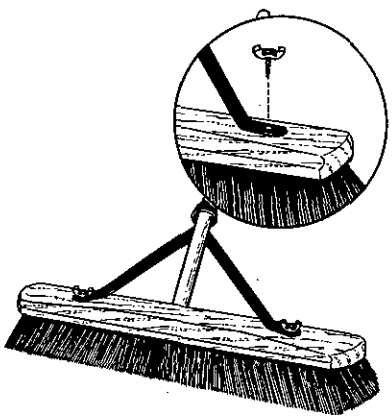


Figure 1B
Tension type



Type II

Figure 2
Strut type





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